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INSTITUTE FOR COMMUNITY INCLUSION - UMASS BOSTON
FALL 2019 RESEARCH SUMMIT
OCTOBER 10, 2019, 1:00 P.M.

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>> This is Rebecca Lazo with Think College and I will go ahead and put two links in the chat box. One is for the live captioning that we will be doing for today's event and one is a link the resources for today's webinar. So the slide presentation and the recording will be there tomorrow and the two handouts that accompany each presentation. And I guess we will get started in a couple of minutes.

>> Rebecca, should we go ahead and get started?

>> Yes.

>> Well, welcome to everyone today. We are recording and that's great. Thank you for joining us today for our Think College fall 2019 research summit. Focused on using the RSA911 data sets and for our understanding of higher education for students with intellectual disabilities. My name is Claire and I work for Think College and I am joined by Rebecca Lazo who also is from Think College who is helping with this event today. We are live captioning this webinar. You can view the captioning using the URL that Rebecca has put in the chat box. It's there if you need it. This is our sixth research summit to date. It feels like we have done so many of these. If you missed any of the past ones you can access the recordings on our website. And we have another great topic we will dive into today with our researchers

sharing their studies that analyze data from the RSA and 911 data sets.

The PowerPoint and handouts are on our website. Rebecca will be putting the link in the chat. We've got some good handouts for each presentation that are also in our Think College fast fact collection. If you are looking for a summary to refer back to after each presentation, then I encourage you to head over to -- I encourage you to head to our website to download those files. You should see a chat panel somewhere in the Zoom controls on your screen. You can post messages to the group in the chat and ask general questions. Rebecca and I will be monitoring the chat throughout the presentations today.

Make sure that you send messages to all panelists and attendees so that everyone can see your question. I do see that someone raised a hand. If you could put that message into the chat if you are looking to get our attention, we would appreciate it.

We have a poll question for you today that -- so we can get a sense of who is here. So please take a minute to answer the question before we get started. Rebecca will put those up on the screen. Thank you.

Since we have a large group we aren't able to have everyone talk without using their microphones so please use the chat panel for any questions you might have and everybody is muted so we shouldn't have to worry about hearing background noise today.

Here are the presentations we have for you today. We have two featured presentations on research studies that were conducted using the RSA911 data set. But before we get to those, we will start with a brief introduction to the RSA-911 data set by Frank Smith, my colleague at the Think College National Coordinating Center at the Institute for Intellectual Inclusion. And will give us foundational knowledge of the data set before we go into our featured presentations.

Let me tell you about the presentations coming up today. In the first of these Tom Sannicandro who is director of the Massachusetts Association on Community Colleges will present his study, effects of college for people with intellectual disabilities on employment, income and SSI. And then Stormy Miller from the College of Marin will present post secondary

education and individuals with intellectual individuals. There will be time to ask the presenters questions after each of their presentations. And then we will have a few minutes at the end to chat as a group about using the RSA-911 data sets to inform our understanding of higher education for students with intellectual disability as well as challenges and opportunities as well as doing this kind of research.

At the end I will explain how you can continue to access support and connect to each other after the event.

First I will hand the control to Frank Smith. Frank manages the evaluation and research team and he has experience with working with large data sets including our tips of data network and the RSA-911. Frank, I'm going to hand over to you to give an introduction to the data set we are focusing on today. Just as soon as the Zoom controls pop up on my screen. There they are. Handing over control now.

>> Thank you, Claire. And good afternoon or morning to everyone joining. Depending where you are. I'm going to set the table here by talking a little bit about the RSA-911 data set so that you have the context you need to understand what our presenters are talking about when they present their research.

So something I don't have on my slide that will be a valuable preface is to remember that RSA-911 data are focused on basically the entirety of what happens for a person who goes through the VR system. So it's not a higher education database. Higher education is something that appears as part of a very large data set. I will explain the context how it gets in there and what we can do with those data.

The RSA-911 -- uh-oh.

>> I think you lost control there. Tell me when to click and I will click.

>> Will do. I think if another person clicks you give up control. Zoom thinks it's doing you a favor but it's not. To get back to this, the RSA-911 is a case service record file that contains info on people who received services from a state VR program. And this is a very large file. It's national. It collects data from every state VR agency as well as DC and the out lying territories and typically including 800,000 records per

file. And it's the -- it's statutorily defined in the rehab act and as it was amended by WIOA so there is a lot of rules built into the collection of the data and the purpose of the data. And the RSA-911 is the federal system so state VR agencies are responsible for maintaining all of their own case record data and then they have to upload it to this federal system on a quarterly basis.

I had a slide in here originally that had the statutory language for what you should -- it is used for and I decided to take that off and you will have to trust me there is a number of statutorily defined purposes of the RSA-911 and those are mainly around reporting on the effectiveness of VR programs and on the cost of services and such so that the rehabilitation services administration and the Department of Education can evaluate what's happening with the programs.

Outside of those statutorily defined purposes, it's a data set that gets used a lot by researchers and practitioners. Researchers use it to better understanding -- understand what's happening in the state VR system and they also like to very often look at a specific geographic sub areas or specific sub populations of VR service recipients to explore what happens when people go through the VR -- a VR program, and how people achieve different outcomes from VR services. And then practitioners like TA providers use RSA data mostly after researchers have published their research to put that information into practice and they use it to improve services and outcomes for people with disabilities who go through the VR system.

So we mentioned that it is a database that includes data on people who go through state VR programs. We need to get a little bit more specific than that. The RSA-911 includes what they refer to as to different types of consumers. First are the participates. These are people -- participants. These are people who applied to a state VR program for services and gotten to the point where they implemented an individualized plan for employment with a VR counselor and have begun receiving VR services. I for years the acronym IPE would get interchanged with IEP when I was writing or communicating with someone. So just remember that, you know, someone might be talking about an IPE and that is a VR system specific document that they are talking about. And participants are included in the calculations of performance indicators that are statutorily defined, and they also collect and report outcome data on these individuals. You

will see the context of why it's important to distinguish those last two columns in a moment.

The other type of people in the RSA-911 are reported individuals. This group is specifically important to people who might be interested in higher education because these are people who have not yet applied for VR services and have thus don't meet the definition of a participant, but it could be students with disabilities receiving pre-ed transition services prior to applying and being determined for eligible for VR services and having an approved IPE. While these individuals are present in the RSA-911 data set, they are not used in the calculations for performance indicators, and the collection and reporting of outcome data is not done for this group of people. But what we do have on them are the other data elements that are in the data set and we will describe that now.

So this unfortunately my footnote got buried a little bit, but I want to preface everything I talked about on this slide saying the RSA-911 case record file has 407 individual data elements in it. Believe it or not, most of those 407 elements are fit under one of the four categories that I list here in bullet points but it's not everything. Then, it's a case record file. So one record equals one individual who interacted with a VR system as a customer. And some of the major data elements are personal or demographic information about VR consumers so the stuff you might expect like age when they started getting services, some demographic information like race or ethnicity or gender, and then personal information such as employment history or education history and sources of income at the time that they first began applying for VR services. Actually, that's kind of goes into our second bullet point there. That baseline information on employment, education and income at the time of application and a big thing that needs to be documented in the RSA-911 is the significant barrier to employment that individual faces that necessitates use of the VR system to receive services.

Then a big portion of the data set are the services received by consumers during participation in a VR program. And so this is a record of all of the services that VR purchases on behalf of the consumer, and there is a very wide range of services that people can receive that get tracked in the RSA-911. It can be things ranging from counseling services about being in the VR system to purchasing educational services which we will discuss momentarily to just all of the other different services

you might provide to someone as you are assisting them in getting employment.

And then towards the end of their participation in the VR program, the participants have information about employment, education and income at the time of exit from the program that kind of gets treated like the outcome information. And I have to confess I didn't double-check in the data dictionary, but for the most recent version of the data set, but typically a person has closed out of the VR system into employment when as a successful closure when they have been in a stable employment situation for around 90 days.

Let's narrow down the big list of data elements. So the way a participation in higher education gets represented in the RSA-911 is in that service data. And so the language can kind of seem a little weird and if you don't have the context for understanding it, you might have a hard time sort of interpreting it and hopefully I can help with that. The way it appears in the data file is that VR agencies can purchase higher education services for VR consumers. And really when they say they are purchasing higher education services, that means that they are paying the cost of the student attending a higher education program. Things like supports or other things that might help a student succeed in higher education like -- would get tracked as one of the other services that VR purchases. And there is data elements to track whether these purchases were made at graduate level, at a two year or technical college. So sort of these different layers. They talk about the education level of the service. Like graduate and technical and two year college. And then each of these they collect when VR officially purchases the higher education services, AKA, pays for the consumer, it tracks who purchased the services, who VR purchased the service from. That's again a nuance thing to interpret. If they contracted with a community rehabilitation provider or some other provider to be that person's case manager, VR will be using that organization as kind of a middle person for designing the services for the consumer. They collect quarterly expenditures on services, and then they do track if and when a credential is earned. And I have a sample of an entry from the RSA-911 data dictionary on the right hand side here. This is for the element is this is four year college or university training provided through VR agency purchase. And I use this one example, but this guidance is the same whether it's graduate level, four year college, two year college or technical college. You can see that

they describe it as full time or part time academic training leading to a baccalaureate degree, a certificate or other recognized less than post graduate educational credential. That *italics* there is my emphasis. Because if you are a state VR agency, you might be looking at this language and thinking to yourself, if we are going to purchase this service, it must be leading to one of the credentials listed.

Then the data go on to say that training could be provided by a four year college, university, technical. So this is how it appears for each one of the levels for higher education in the data dictionary.

And then there have been a lot of recent changes to the RSA-911. I started with it in 2005 and what is available now is very different. So just in case anyone read research in the past and some of the research we talked about here will be reflective of the old structure of the data set. It used to be a data set that was just for people who exited the VR system. So people are not input into the RSA-911 until they exited a VR program. Now it's anyone who is currently on the VR rolls, so to speak, which is why there is a larger number of records in the data files than there used to be. I typically used to work around 500 to 600,000 records per year and now we are over 800,000.

Other recent changes is that it's available on a quarterly basis. It used to be provided on an annual basis. Then there are changes to how and when employment outcomes are reported. This was done to make the outcome data better align with what the VR programs need to report for their common measures, so things like wages two quarters after exit as opposed to 90 days or another increment of time. Then a notable addition to the list of services you can report on is work based learning experience. I don't have a definition here for how they describe that, so prior to the new iterations of the RSA-911, you could not report on a service called work based learning.

This leads to accessing the RSA-911. It's managed by a group at the Department of Education, and the first step is just to e-mail them at the address that's listed on this slide right here and just tell them that you want a copy of the RSA-911 data file. And you will have to sign a formal data use agreement with the RSA in order to get a copy of the data file. And once you do, what will get delivered to you is what's known as an ASCII file. So you are getting a massive text file that is organized

into columns of data and rows of data, and in order to convert that into something that you can analyze, you will need to use some sort of tool that allows you to write code, to read all of that information into a case record layout, whether it's an Excel. Is there anyone from RSA on the call? Maybe they wouldn't speak up if they were, but I will just say that it's very daunting and very difficult to make that data file out of an ASCII file, but we've worked with people in the past who have had data use agreements and if an organization has an SPSS or Excel version of the RSA-911 that they prepared and have -- haven't manipulated any of the data or, you know, they don't have any columns in that data that you aren't supposed to have as a researcher, then as long as it's a data use agreement, sometimes you can convince an organization that has a prepared copy of the data to share it with you and that's not really a problem.

I will be happy to take some questions and I will monitor the chat.

>> Yeah, I think -- thank you, Frank. That was a great overview of the data set. I think given the time we should go into our first presentation. But certainly if you have any questions about anything that Frank talked about will go in the chat and we will get to those. Thank you so much, Frank. That set the stage for today.

So next I would like to welcome our first presenter -- or first featured presenter, Dr. Tom Sannicandro. Tom is the director of the Massachusetts associate of community colleges of before that the director of the Institute for Community Inclusion at UMASS Boston and served 12 years as a state representative in the Massachusetts legislature. In his work as a state representative, Tom served as the chair at the joint committee on higher education and sponsored a number of ground breaking initiatives including the enrollment program that enables students with intellectual disabilities to attend college in Massachusetts. Tom is the father of an adult son with Down Syndrome. He earned his PhD and I have to say he is the only one of our presenters today or so far to have a Wikipedia page which made it easy for me. So over to you, Tom. Let me just turn off my video and then I will hand control over to you.

>> Great. I think I have control now because it says that I have that. Thank you, Claire. I'm excited to present this research. This is the first time that I have ever gotten to present this so

I'm excited. And I think the findings are important especially important for folks that work in the area of providing services or education for individuals with intellectual disabilities and those of us who look at policy and try to figure out how do we get these individuals to have the best life that they can have.

Now I understand what Frank was talking about. There we go.

I will go briefly what I will do just a little introduction about why I did this research, what the aims of the research were. I will walk through the research method, talk about the findings and then talk about the strengths and limitations as well as the policy implications, and then future research. Some of that we may be covering later on in the question and answer period.

First off I think is why this research? Those of you on the call probably are very involved with the work at Think College or you wouldn't be here, but the real issue was how do we best serve individuals with intellectual disabilities? In this country we have three to six million people that would fit into this category. What is the best way for us to provide opportunity for them? We know in the K12 system with the special education law that everyone has a right for an education if they have a disability, but that opportunity only exists in our second -- as far as our secondary schools. There is no guarantee of higher education for someone with a disability. As matter of fact, there are a lot of barriers that exist for someone with an intellectual disability for attending higher education.

Then what's the best way that we can prepare students? I think one of the things that we looked at and that I looked at in this research and one of the things that Think College is continually looking at is employment outcomes for individuals. And you know, in my mind that's not the only reason that someone should get an education. It's not all about employment, but that's one of the major things that I was looking at in this research. And what difference does it make if you do have higher education.

The historical context we all know for folks with intellectual disabilities -- this has been just a -- even in the last few hundred years has been really particularly horrible for people with intellectual disabilities. In the 1970s, we saw the

closing of large state institutions and that's when we first saw the beginning of programs on college campuses. Probably wasn't until 2008 with the changes in the reauthorization of Higher Education Act that we saw real dramatic changes in the ability or the availability of college programs and the growth of college programs for individuals with intellectual disabilities across the country. And across the world for that matter. We are seeing them really across the world. The employment rate for individuals with intellectual disabilities is 23% or at least was at the time that I did this research. And employment rate is the percentage of individuals who are actually employed. There is a controversy or a question about using unemployment rates. One of the challenges is if you look at 23% of individuals with intellectual disabilities are employed, and you look at the disabilities as a whole, 30% of them are employed. And then compared to non-disabled people, 79% are employed. So we are looking at -- for individuals with intellectual disabilities. One in three are employed as compared to someone without a disability which is absolutely dramatic.

Frank talked about the vocational rehabilitation program so I won't spend time on that. And I won't get into the literature review because we will just talk about what I looked at and what we discovered at this point.

The primary aim of the research was to say what was the effect of post secondary education on employment for individuals with intellectual disabilities? And the way I actually posited the data that I looked at did the individual have some college, and did they have some college experience? I didn't look at training per se and there is training around the area of sort of the soft skills kind of training and those types of things. I was really looking at in the data set there is -- you can parse it out so that did the individual have some college. So that's what I was looking at. Determine did they -- did students with intellectual disabilities, did they benefit from attending post secondary education. Different from those who did not. Then I dug deeply into how the VR agencies were treating individuals with intellectual disabilities as far as the number of folks they were bringing into their system and offering services to because states vary. Some states -- for instance, Massachusetts really discourages people with intellectual disabilities from entering the VR system and you can see disparities of people across the country if they bring people with disabilities into the system or not and there are different reasons why it varies and I looked at

that as well. I also looked at the time period I was looking at was 2008 to 2013 and I looked generally what was the growth of programs for individuals with intellectual disabilities in higher ed in that time period.

The data set I used I used the RSA-911 that Frank talked about. I looked at when I started the research, I only looked at individuals age 16 to 30. And then I only looked at fiscal year 2008 through 2013. And this will give you a sense of the numbers that we are dealing with. When you look at that between that time period the number of observations that were in the data set were 3.6 million. I only looked at individuals with intellectual disabilities. So I dropped everybody out and so then I was looking at 300,000 individuals. Then when I dropped out, I only looked at individuals with ages 16 and 30. I ended up with 180,000 individuals with intellectual disabilities that met all of those criteria. And I'm going to get into and talk about the propensity score. The propensity score is essentially creating a comparison group of like individuals because I think the biggest thing with the research that I did was that I created a comparison group because one of the questions that I had was does college make a difference for these individuals? Or were they already pre-destined because of family supports? Their ability to learn thing in an academic setting. Whatever it was, was this group more likely to get a job for the same reason they were more likely to get into higher education? Or was it the higher education itself that caused the better employment and better other outcomes. I'm not sure. What I did was parse this group of 180,000, created a comparison group and I will talk about that briefly in the statistical analysis. When I did that I only got two groups -- I needed two groups that were exactly the same on these criteria. When I did that, each group was about 5500 people. So I ended up only analyzing 11,000 individuals over that -- the six year period.

I also pulled in other data as well as the RSA-911. I lumped other data in so I could look at things differently. I looked in whether they are employment first state. I looked at how much the DD agency in the state was spending. And then I pulled in state economic data looking at employment rates for individuals with disabilities as well as those without disabilities and then I looked at poverty levels for the state to just sort of look at what is happening.

As far as dependent variables I used. I was looking

whether they were employed, and that's when the VR case was closed. Did they end up with employment or not? What were their weekly earnings, with what was the amount of their SSI benefit per month and then I looked at what was their occupation at closure? What I was trying to figure out, did these individuals go through this training -- go through education and then end up in the same types of jobs? Or did they get different -- I live by the train tracks as you can tell. Did the education allow them to get jobs that were not typically held by someone with an intellectual disability? That was another issue or point that I looked at.

As far as independent variable, I was looking at post secondary education and really I was looking at college. So it was really college that was being examined in this point as opposed to secondary education as opposed to some other post secondary education and then looked at the amount of money that the state VR -- the state Voc. rehab agency was spending per medium income in that state to standardize that across the country. I also covariant gender, race, and the other training provided and that's like the soft skills stuff that you can get. How much the services were that the VR agency purchases for that individual and were they a Medicaid recipient at time of the application? And using that covariant, I was trying to get issues of poverty is what I was looking at and that's why I pulled and used that particular covariant.

>> Tom, we have one question for you. The 2000. To 2016 time frame, why?

>> 2013. Because I actually did this -- when I was getting my doctorate and that was the last six years that were available. One of the things in my overall research I won't get to but it was the period of the great recession. It even complicated things even more which is interesting, but it was -- there was no reason I guess is the easy answer. It was the time period that I was in and the data that was available when I did my research.

>> That's great. Thank you.

>> Thank you.

>> The research design was a quantitative analysis of the secondary data set is what we are all doing. The other piece which I talked about before was I created a comparison group and

then I also did a hierarchical linear model or logistic modeling. And for those of you, you probably -- a lot of you know what I'm talking about but essentially you look at states as the individual and then the state that they fall under that why you do a hierarchical linear model if you can think about it, if you are thinking about unemployment rates for someone with an intellectual disability, if the unemployment rate is say, 5% in Massachusetts and the unemployment rate in say in Florida is 10%, that is going to cause a big difference among what's -- among your results. Or if you get people getting employed in Massachusetts at a higher rate and then they are employed in Florida at a much lower rate, when you average it all together, it's blurring your results. So you really can't tell if anything happened. So what I did is I used the hierarchical linear model to get rid of all of the variance of the states to see if is there something happening here so that we don't lose the data that is happening as we look at the sort of nuance among the states.

This is where I get into the comparison group. In my opinion this is the importance of what this research looked at and what I did was I came up with four items, four variables that really indicated whether a person was going to get a job or not. There is one of the variables in the data set is significant disability. Another one is did the individual receive special education services in their high school experience? And whether the individual was employed at application and the level of education at application. So what I was trying to do was create two like groups. So that we know that -- or it seems to indicate that the college experience is what's causing the difference is the point at this point.

So when I looked at these, you can see the propensity score when I looked at the issues and you can see the difference here on the test statistic, when I looked at this first column here, they have an intellectual disability but they didn't have post secondary education. The second column is they had an intellectual disability and they did have a post secondary education. You can look at the test statistic here. These are dramatically different populations. So there is a big difference between the first column whether you had post secondary education or not. So that was the problem that I was trying to get rid of because these groups are not the same in anyway and statistically they are very, very different. So what I did was I standardize -- essentially standardized, I used data and used

nearest neighbor matching is the method to do this. I standardized essentially on these four points that appeared to be through -- I did some other digging through the numbers to see what really affected someone's ability to get employment. These factors were really critical in that process and that's why I chose these four factors to standardize on.

After I did the propensity score matching, I essentially got rid of the difference. So now we have two groups that are identical on these four issues. Both groups have intellectual disabilities, but 5,000 of them or 5500 had no post secondary education. And the other group of 5,000 or 5500 did have post secondary education. So essentially they are identical on these standards.

These are the findings and this is using odds ratio. They were twice as likely to be employed if they had post secondary education which I think is a dramatic finding particularly in the light of the fact that these are two identical groups on factors that would indicate that they would be likely to be employed. The odds ratio of getting a non-traditional job meaning non-traditional for someone who has an intellectual disability was four times greater. And what gets pulled out of the non-traditional job with the service industry and those types of things. All of those factors of traditional jobs or there is a job item in there that Frank had talked about which is SOC which I don't remember exactly what it stands for, I'm sure some of you on the call know that, but that's the standard that they used in that system.

The earning per month, they earned -- I think it's like \$400 more per month, approximately, maybe a little less. And they relied about \$100 less per month on -- on the income. So the earning more and they relied less on government benefits. And this is the hard crunching data that you can look at. The only thing that I think is -- the gender issue, this finding which I probably should have done a different way, the finding is to African-Americans. So the comparison between someone who an African-American individual who had an intellectual disability and post secondary education, there is a dramatic difference between the white individual who with the same -- with everything else being equal. So this sort of racial prejudice and whatever is causing this issue, the and -- prejudices and systematic racism that exists in the United States does not spare and does not change for folks that had -- in this study and we can see

this as we go through.

Also the cost of services had a dramatic influence. Of what it was done. Employment was a big deal as well as the higher the number of folks that are served in the VR system with intellectual disabilities, the better off the results are.

>> Excuse me a second. This is Rebecca. There is a question in the chat asking for examples of non-traditional jobs. I don't know if you have that information available.

>> It would be a management, more of a management professional job. Like a service job would be someone serving food in a catterria. A management job -- cafeteria or a management job is a job where it's more of a professional kind of job.

This just tracked the rise of individuals with intellectual disabilities in post secondary education. So we are seeing a rise as we go through the process. As we go through the years and obviously Think College will have much more information on that. The strengths is this was a national -- a big national data set we used other variables from other data sets. The biggest strength I think is the creation of the comparison group that said look, let's figure out what's going on here. Selection bias I think is a big issue but I was hoping to get rid of that with the comparison group. The other challenge I had was that the identification of a post secondary -- post secondary education and the identification of someone with an intellectual disability. And I thought that was a big -- I'm not really exactly sure whether that was always accurate or not. Obviously the folks at VR agencies are saying it is, but I was quite concerned about that. And also the issue around selection bias, whether in fact there is a VR system or not. Because I only looked at individuals in the VR system. So there is a selection bias that exists in this, too. And then you can come in and out of the VR system and if you continue to come in and out you get reported as a new observation the next year you come in which is obviously creates a problem.

For future research, I could only look at did they have some college or not. So how much college they had and how much of that made a difference. I can't tell from this data set to look at other populations that aren't in the VR system or more in-depth qualitative studies that would figure out what's happening. And I think the biggest thing that says that we need

to be thinking about special education in the role of college for individuals with intellectual disabilities as well as looking at Voc. rehab agencies and are they serving who are they serving and what that looks like and then the reallocation of resources -- from these results it appears we should be putting more money into college for individuals with intellectual disabilities. That's it!

>> Thank you so much, Tom. I think we did have a couple of questions that came in while you were presenting. They are hiding from me because you were controlling my screen.

>> Sorry.

>> Could you ask that question from Cathy? If you are there.

>> I'm here. I just have to find it. Cathy asked about -- she said don't mean people with ID get medicate -- Medicaid from disability rather than poverty.

>> Yes. So you also, though, since we started, I did 16 through 30, so you would also get some that were not -- that did not receive it. And there is no way to look at the poverty issue so you are absolutely right, but that was why I did that. So it may be skewed, but it was something that I think allowed me to look at that issue.

>> Yeah, she also sort of noted wouldn't the fact that many individuals with ID being on Medicaid confound the issue because they end up being people of poverty -- sort of a --

>> Right.

>> And I think that was it.

>> Great.

>> So if anyone has questions you can put them in the chat or something before we switch over.

>> Thank you for reminding us that SOC is a standard code. Thank you for your presentation and we will head over to -- Cathy was asking if your study is published.

>> Yes, it's in the -- I can give it to you. It's actually been

published in -- and I can't remember where it's published. The whole study is my dissertation. I will send you the link later, Clare.

>> And I believe if you go to the website and download the fast fact that Tom wrote for us, I think at the end of that, there is a reference --

>> Oh, there is?

>> Think I so. I'm sure.

>> Great. Thank you.

>> Next I will introduce our next two presenters. Dr. Stormy Miller Sabia is director of student services and psychological services at the College of Marin where she worked since 2017. She earned her doctorate of education with a concentration in post secondary education and community college leadership from San Diego State University. Prior to assuming her role at the College of Marin, Stormy worked in disability support programs and services at the San Diego community college district that has more than ten years of experience in the field of disability in the community college and non-profit setting. Her research focuses on the association between certificate or degree attainment and earnings among individuals with intellectual and developmental disabilities served by vocational rehabilitation.

Dr. Mark Tucker is associate professor in the Department of Administration, rehabilitation and post secondary education at San Diego State University. Mark became a full time faculty member in 2014 after spending 14 years with the into work institute at SCSU. Mark earned his PhD from the University of Northern Colorado and is a certified rehabilitation counselor. His areas of research include examining the association between post secondary education and training and vocational rehabilitation outcomes as well as the relationships between individual contextual and case service factors and rehabilitation outcomes of transition age youth and adults with disabilities. If you have any questions for stormy and Mark during their presentation, use the Q&A or the chat box. Right now I am going to hand over to Stormy and Mark. Give me one second to give you control here.

>> Looks like I have control now.

>> Great.

>> Thank you. Second, Clare and Rebecca and Frank and the entire Think College team and for those who are joining us on today's webinar. It is a pleasure to have this opportunity to be a part of the fall 2019 research summit and to share our work using RSA-911 to inform, understanding of higher education pursuits of intellectual disabilities. And then I will preface this saying that it's my sincere hope with all of the -- with all of the power shut off in California I was able to be with you. One spot that internet was going out but I think we are okay.

Just provide a little bit of context. There has been great deal of momentum in development of policy particularly in California for a number of years now with our previous governor signing the California employment first policy making employment a priority among working age individuals with developmental disabilities. There is also the California blue print for competitive and integrated employment which was a joint effort between the California Department of Education, Department of Rehabilitation, Voc. rehab and the Department of Developmental services. In addition, permitting customized services in areas such as employment for secondary education for individuals with intellectual developmental disabilities certified with developmental services system.

So given the legislative impetus of expanding opportunities for those with intellectual disabilities and increasing the need for the college experience, associate degree or a baccalaureate to meet industry expectations is an incredible opportunity to explore post secondary education service provision by students who -- with particular attention to California vocational rehabilitation.

So Frank did a wonderful job in terms of providing context on RSA-911 and Voc. rehab as a refresher. It's a system designed to assist college individuals with disabilities with attaining or retaining employment and improve employment outcomes is one of the largest federal programs that can support employment outcomes among individuals with disabilities. And to assist individuals with their employment goals and then as was mentioned, the Voc. rehab ask provide financial and other sources of support for those who are pursuing post secondary education training. This might just take a second to move to the next slide.

So we also touched on this during Tom's presentation, but many are aware of the disparities between individuals with and without disabilities and labor force participation and employment in earnings as well as the underrepresentation among individuals with disabilities and post secondary education and degree completion. They are greater disparities among individuals with intellectual disability in areas of preparation for work, after high school employment and earnings, and individuals with intellectual disability are less likely to have goals related to education and employment reflective in transition plans and I think this goes back to some of what Frank was mentioning in terms of we get the individual life plan for employment which is developed by vocational rehabilitation and then individualized education plan IEP that would have the transition plan, but speaking specifically about the IEP and how there may be less goals that peek to post secondary and employment. Although there is a clear link between education and earnings potential.

So the purpose of this study potentially it was examine the status of individuals with intellectual and developmental disabilities who receive education and employment support through vocational rehabilitation. Examination of the association between post secondary and completion in earnings, also looking at the similarities and discrepancies in which the earnings which the individual with those who completed college and then really looking at trends and sort of profession of post secondary ed support by vocational rehabilitation. And as I mentioned earlier we were specifically looking at service provision in California and comparing California with the research questions to all other VR state agencies through the analysis of the record using RSA-911 and I think again it's something that was mentioned in terms of at the time of this study we were looking at there is that expansion and the case records and this is looking primarily at case records at closure.

Just to provide a bit more of a California landscape, spoke a lot about the history of implementing legislation and policies in support of post secondary education and employment. But also wanted to touch on the California VR agency. Which has 1800 staff and then serving close to 100,000 individuals and this is looking at federal year 2014. More than 36,000 of the 100,000 were new applications. This also had -- California has a robust higher education system. There are 115 California community colleges. There are 23 colleges and some of the largest

university systems in the nation and then there are ten University of California campuses that serve approximately 250,000 students. This is just touching on public higher ed in California. You see on the slide that this 2015-16 looking at the budget of 29.2 have been allocated to California higher education and look at 1920, that's 36.9 billion. So pretty significant increase in terms of what is being allocated in higher education.

Mark, did you want to add to -- regarding the California landscape?

>> I'm not sure there is a whole lot I would add. Just with respect to talking about the California Department of Rehabilitation is a large agency. Other state agencies might have total staff in the hundreds or maybe 100 in some cases in small states. Relatively large number of folks also covering a relatively large population.

>> So going into the message for this particular study. There are four main research questions which we will share with you utilizing RSA-911 data set to assess the provision of post secondary ed. The differences in earnings among individuals with intellectual and developmental disability who did and those who did not complete certificate and degree. And also looking at the differences in earnings among those with intellectual and developmental disabilities who did complete college while their case was opened with vocational rehabilitation. We specifically looked at several fiscal years 2006-16 and two of the research questions were intentionally developed to look at trends and patterns in the provision of post secondary education to individuals with intellectual and developmental disabilities. And to design intellectual and developmental disability we were looking at the American association on intellectual and developmental disabilities -- [inaudible].

Did I jump too far ahead? So the research questions. And I will go ahead and read them. So the first question was there a difference in mean weekly earnings at case closure and we are looking at individuals with intellectual and developmental disabilities in California who advanced to post secondary certificate or degree completion while their rehab case was open and individuals with intellectual and developmental disabilities who did not advance to post secondary certificate or degree completion. The second research question we were looking at any

significant differences in the mean weekly earnings at case closure between individuals with four different types of intellectual developmental disabilities in California, so cerebral palsy, autism, epilepsy or intellectual disability. So this is among those who did advance to certificate or degree completion.

Then the third research question, when examining the provision of post secondary education support by the California Voc. rehab agency between the federal fiscal years 2006 and 2016 were there any significant year to year differences among the proportion of rehabilitation cases provided with college for individuals with intellectual and developmental disabilities. And then the last question we were asking, looking at case closures during specifically looking at federal fiscal year 2014, were there any significant difference between the proportion of individuals with intellectual disabilities who were provided with college when comparing California vocational rehabilitation to all other vocational rehabilitation agencies combined.

And just with regard to data analysis there were several methods of analysis that were used to investigate the research question. For independent sample with -- to examine the differences between mean weekly earnings between individuals who did or did not advance to degree or certificate rebill -- completion while their cases were open. And then the career test was used to investigate year to year differences among proportion of rehabilitation cases provided with college or university training compared to those looking at individuals of intellectual disability to those that did not have a primary diagnosis and then -- independence with performing to look at their significant differences in provision between California and all other VR agencies specifically the -- the post secondary education.

Mark, is there anything you want to add there?

>> Just, with respect to the third question where we looked at the provision of post secondary ed support by VR agencies over time, I guess there are two things. One is initially the study period we looked at was 2006 to 2014 and then anticipation of this particular presentation we took a look at additional two years to get a more up to date perspective there. So that's one thing. And then the second was that the way we constructed that question was kind of in line with something that Tom had mentioned earlier which was a concern for given fluctuating state

budgets and things like the great recession and stuff like that, you could see agencies having different attitudes about finding expensive service like post secondary ed over time and in order to control for that the way we looked at it was not like the total number of cases or the total percentage of cases that were provided with post secondary education support but we were looking at it from -- within each year what proportion within that year of all of the cases provided with post secondary support were cases for individuals with a primary cause of impairment of IDD. And in that case it's a recession where we are spending on post secondary ed might drop, we are still comparing during that year of all of the cases that did get funding what proportions were IDs -- or folks with IDD. In that respect we tried to construct the question to control for variations in -- for general variations in funding and focus hopefully that would allow us to focus in more narrowly on our particular question which was: The folks with IDD provided with post secondary ed support through VR.

>> Okay, so the results. -- oh, I went too far. Okay. So this is looking at research question number one and so we are looking at those who completed their certificate or degree. Specifically looking at individuals with intellectual developmental disabilities and then also looking versus those who did not complete and did not advance. So there were 8,722 cases closed but did not advance in level of education and 247 that did advance to college completion when their case was open. And there was a significant difference between the mean weekly earnings between the two groups and I believe the difference was \$146 in mean weekly earnings.

>> Just to add to that in addition to as you would expect a difference in earnings of that side, we took a look at effect size which is a measure if there is a significant difference how big is that difference and they get categories of like small, medium or large and that was a large effect size. So not only is it a significant difference but the difference between the two groups is fairly large.

>> So we are going to come back to the second where we are looking at the differences in earnings between those with intellectual disabilities and those with developmental disabilities that did advance and going to the third and looking at the proportion of the individuals who were provided college, the provision of the secondary support by Voc. rehab who did have

intellectual developmental disabilities. So again as Mark mentioned initially we were looking at that 2006-2014 federal fiscal years and then expanded that. So there was significant year to year differences and you will note that there are a few years where there were not in that 2007, 12, 13. And in 2006, 8, 9, 10 and 11 proportionate rehabilitation cases were lower than the expected count and that 14, 15, 16 year the proportion of rehabilitation cases in college among the individuals of intellectual developmental disability was significantly higher than the expected count. And one of the things that one can't be sure that is thinking back to some of the earlier remarks regarding the higher education opportunity that was thinking about policy development, certainly with the 8, 9, 10, 11 the great recession is thinking about what was going on there. But start to wonder if policy perhaps influenced decisions in terms of funding in that -- one cannot say but it's something that we found interesting.

Mark, did you want to add anything here?

>> Just with respect to interpreting the graphic you are looking at, you know, what we will think about if you were to look at 2006, what's being answered, the question being answered there is that of all of the cases closed in 2006, what proportion of cases were people that had a primary cause of impairment of IDD and it's around 4.1% and so then if you moved to 2007 that's about 4.5% so it's kind of a slight increase and as we get towards the end there, the ones that are green where the portions are significantly higher, you know, those proportions went up considerably so in 2014 of all of the cases that were closed that year where post secondary ed support was provided. 6.5% of those cases were people with IDD and in 2018 it was 6.8% and 2016 it was 6.9% of all of the cases that were provided with post secondary ed support with people with IDD. For us looking forward one of the things we are curious to see is if this trend continue or is this kind of a short-term thing.

>> And so looking at the research question number 4, so this is where we were looking at the nationwide proportion -- nationwide proportion of individuals with intellectual and developmental disabilities provided college compared with California and it was specifically looking at federal fiscal year 2014. And here there was a significant difference in the proportion of cases provided with college. I believe in total -- including California was 3.2% where there was a provision of college the secondary and

then looking at California it was 7.2% of the rehabilitation cases where -- that were provided the provision of post secondary ed compared to 2.9% for other VR agencies combined. So that was a difference in terms of the portion of cases that were provided college. Anything else there?

>> Yeah.

Anything else there?

>> Just to clarify things in the previous example we were comparing people comparing the proportions of folks with IDD and folks without IDD that were provided with post secondary ed. In this case we are looking at of all cases of people with IDD what proportion of those cases, 100% of cases with IDD were provided with post secondary ed support. So in this case California was providing post secondary ed support to a greater proportion than all other states and territories combined as a whole. One thing to keep in mind is that in California even that 7.2% that was meaning that post secondary ed support was provided 291 out of the 4,054 cases of individuals without IDD. While it's a greater percentage it's a fairly small overall number.

>> We have one question from one of the attendees asking what was the -- this is a slide or two ago. What was the expected count you were referring to? I think that was two slides back.

>> The expected count would be kind of like the average proportion across the whole -- what is that? 2006 to 2000 -- a whole 11 year period. So if the proportions were on average around 4.3%, if it was lower than that, significantly lower than that, then that's when Stormy would say the post secondary ed support was provided to a lower count or significantly lower proportion of individuals than would be -- and the same thing with higher. In an instance where the average cost was 11 years was around 4.3% a year where post secondary ed support is provided to like 7.3% and that would be a significant higher proportion or a significantly higher count.

>> Thank you.

>> Mark, do you want to dive into the second research question if we go back to that we were looking at individuals of intellectual and developmental disability that were provided post seconded support Voc. rehab and did advance in certificate degree

completion but we were looking at the mean weekly earnings between specifically looking at individuals with intellectual and developmental disabilities.

>> Right, yeah. So we wanted to see for those who did advance while their case is open their level of education increased and got post secondary ed support and their level of education increased between the lower application at application and at closure. Of those folks who already a select group of folks kind of distinguished by their educational achievements, we wanted to know the ones who are successfully closed, was there a significant difference in the mean earnings between the four groups. ID, autism, cerebral palsy, epilepsy. And the answer was basically no. So there was no significant difference between their earnings. Looking at the mean earnings, they did vary quite a bit in terms of the averages. The differences if weekly earnings were sizable, but it was also a lot of variation within the groups as well. And a lot of variation and earnings within cerebral palsy within ID, within epilepsy, cerebral palsy and we tried throwing out the outliers and the results were non-significant. So the second question with the data that we had no significant differences between the mean weekly earnings of those who advance their highest level of education when you compare the four groups kind of assume the IDD label.

>> Thank you. So implications for practice. There does seem to be an opportunity for individuals with intellectual and developmental disability and exploration of the secondary education as a support as part of that individualized plan for employment development through vocational rehabilitation. Even looking at use of intellectual and developmental disability planning could be done as I mentioned before thinking about now talking about the individual life education plan during that transition process is there is opportunity to build in those goals around post secondary education and employment before that would help them actualize that goal of VR support. Really thinking back to the preference of work base learning and we are looking at -- the rehabilitation not -- not just California but thinking about what we call the Department of Rehabilitation student services which gives that opportunity for transition age 15 to 21 to be able to access some work base learning opportunities as long as they have an individualized education plan. So is there an opportunity to be able to realize that goal through at -- development.

There is movement occurring and not specific to California nationally but thinking about that program shift with K through 12 and looking at vocational rehabilitation in other developmental service organizations in California. We have the largest Department of Developmental services in the regional center system. So the thinking about that system here. We are looking at the way that they partner but is there opportunity to extend to community colleges and four years and local partnership agreements. We have some of that going on but there is still a lot to be developed in that particular area. So it's not already in place and that may be a real opportunity in terms of practice.

Particularly given the energy around the development of post secondary education programs, specifically designed to serve individuals and intellectual -- [inaudible] disabilities. We hope to continue to see policy development in the arena of education access for individuals -- [inaudible] and as policy develops, speaking to the way that colleges can play a vital role in education in work force preparation. And again I referenced that vocational rehabilitation partner rehab student services. So would there be potential for colleges to provide those services in preparation for -- [inaudible] also looking specifically at the provision of post secondary and support by Voc. rehab, but how do we with all of the designation that Frank shared in terms of who -- the post secondary support ed is provided but is the provider involved or other supports that are also assisting in post secondary ed support. So how do you blend funding? How do we pull resources to be able to further provide access to post secondary education? And again this goes back to collaboration and communication as it relates to funding. It could be done with local partnership agreements.

And also thinking about how are we educating use with intellectual and developmental ability and family members so that post secondary education is as a viable option for those who desire it and also try to navigate what can be a relatively complex system as you start to think about the statewide agencies and other organizations that are involved in the provision of support from a funding aspect but then also to address the service.

Mark, is there anything else you want to add in terms of the implication of practice?

>> No. I mean, I feel like you definitely covered the landscape

well there.

>> So future research. With regard to future research, there is opportunity to examine other potential systems both secondary and support and I think we will probably be touching on the limitations, but again I mentioned in California that might look like our Department of Developmental services and the regional center system as one potential opportunity. We might also be looking at work force development and employment development and department of. There were some work at the end of the study there were a lot of -- just winding down you start to ramp up again thinking about the possibilities given the new data elements with RSA-911 so certainly there are feelings that there is potential here to look at zip codes and then also examination of wages based on industry specific areas. So that was something I just really feel like that might be that next steps really narrowing it down even more and not just California as a whole but looking at specific areas. Then perhaps even comparing communities to comparable size and communities for unemployment rate and industry opportunities.

There are also opportunities looking at that goes back to thinking about those student services, work force innovation opportunity act, we are looking at transition. 16 to 21 and the application may be shift in services. Would there be a shift in case duration as well. Are they open longer and you are taking into account typically you would see that specific area where it may be open longer, but specifically looking at age at application and playing around with some of those new services and elements that you might want to take into consideration.

Limitations, the big one is it's hard to -- really operating within the context of what we are provided with RSA-911. A massive data set but ultimately you are not going to be able to get particular things out of it. So the limitation was addressing research questions beyond the scope of the data set. There is also the -- even though there is measures put in place to try to limit the possibility there is always a possibility of incorrect input of information because it's human error. In terms of what's being put into the data that's being collected. The outcomes also this particular study we can't generalize that to those who do not -- rehabilitation services. We can't speak to individuals with intellectual and developmental for secondary as report from other systems and what those mean weekly earnings might look like for that provision of support

might look at. May not specifically be listed in terms of RSA-911 at the time that we were doing the study.

And thinking about conclusions here, wrapping up, I think it provides information in the studies, Toms as well as this one, providing information on the current landscape and potential shifts in service delivery and thinking about how we can increase access and how specifically how vocational rehabilitation can increase access for the provision of post secondary work. Raises awareness of options from policy, practice, research and then again really looking at post secondary that is greater and more than an opportunity specifically for individuals with intellectual and developmental disabilities part of our overall higher education initiative. And I think that's it.

>> Thank you so much, Stormy and Tom. I will take back control here. I think we answered the questions that have come in along the way. And so thank you Rebecca for handling those. Amy Armstrong point out that the data from both of these studies are really helpful if we look at program sustainability and that's one thing that I really liked about both of these studies, especially I think both of them started a dissertation work and so then going to be a published article and then it's great that you both did that which means it's out there more widely available for those of us who have to write things every once in awhile to advocate for continuing access to higher education for students with intellectual disability. I think it's interesting these independent studies using the same data set reached the same conclusion that this is an effective model and we need to make sure that we are continuing to provide access to -- [inaudible] and enabling access to higher education.

I don't think there are any other questions specifically about the studies. We will have a couple of minutes -- let me ask both of you from a research perspective if there are any challenges that you encountered when working with a large data set like RSA-911 or perhaps some lessons learned doing that kind of research. Mark and Stormy, I see you laughing. What is your perspective on that?

>> Stormy, do you want to answer?

>> I will let you take the lead on this one.

>> So I think and I expect Tom had similar experience with this

as I know you obviously stitched together data from a number of years as well. That's a pretty phenomenal undertaking itself bringing together data from ten or 11 years worth of program operations and you get millions of cases at that point and then drill down to the ones that are of interest to you. Not a static data set from year to year. As Frank talked about early on, it's now transitioned to not consisting solely of closed cases. And it's issued quarterly now and has a lot of other variables in it that weren't in in it in the earlier years and even in the earlier years when it was closed cases and only closed cases in whole year chunks variables change from year to year. So some the variables we were interested in in tracking they are defined a little bit differently over that period of time. You have to make a lot of judgments about is this okay to keep -- or do we need to give up on this because the variables have changed sufficiently -- changed sufficiently that we are not measuring the same thing so we ran into issues like that that variables changed, definitions changed. That in itself -- we spent a good chunk of the data analysis and the data -- just all of our time with the data just forming the data and getting it ready for analysis.

>> How about you, Tom?

>> Well, I agree. That was a major problem. Sort of cleaning the data and getting it ready and lining up -- it wasn't so bad but it was some slight changes in the variables in the years that I looked at. I think the other piece that I have a challenge with data itself where I had a coal colon in the wrong place and I thought for months and months until I realized if was the wrong place and then I fixed it and then it ran. Those are some of the problems you run into.

>> Yeah, exactly. Yeah, it's definitely presents its own set of challenges working with a large data set like this one. But, yeah, it's valuable work and great conclusions that we can draw from this research.

It looks like we have one minute left so let me finish up with closing announcements here. Thank you again to all of our presenters today. We really appreciate your time and your dedication to supporting research in this field. If anyone needs a certificate of attendance, please let us know. As I mentioned earlier we have a few resources for you to download from the event page on the website. The PowerPoints and the two handouts.

The recording of this webinar and transcript will be posted on that page. And we would appreciate if you would take a moment to share your thoughts on today's webinar using the polls magically appeared on your screen. Thank you, Rebecca. So we can be sure we are doing our best for you. You just need to click a few times and we will have your feedback. Thank you for that. If you have not already joined our affinity group on research and you are interested in continuing to connect with other people around research in higher education for students with intellectual disabilities I encourage you to do that you can head to our website and see these slides that your poll is in the way, click on the technical assistance and then affinity groups and then the research affinity groups. We have an affinity group on vocational rehabilitation if you are here more for the content today, that's probably a great group for you to join as well.

I hope to continue the conversation with all of you through those groups. If you have an idea for a future research summit or a study you would like to present. Please send me an e-mail. And as a reminder, you can always e-- you can always e-mail us with your questions about inclusive higher education at Think College@gmail.com. One more slide there. So, yeah. Thank you for joining us today. Have a great day.

>> Thanks.

>> Thank you, Mark, thank you, Stormy. And thank you, Frank.